# M13 Batch 1 — Full Set

M13.1 Theory of Flight · Rotary wing · PDF pages 1–63

Terminology and blade geometry, lift and gyroscopic precession, classification of rotary-wing aircraft, the cyclic, collective and anti-torque controls, swash plate and flapping, ground effect and autorotation.

105 questions · Level 3: 51 · Level 2: 34 · Level 1: 20

● marks a question built on a fact your instructor highlighted or underlined in the manual (100 of 105). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

## Terminology and blade geometry

PDF pages 4–20 · 25 questions

### LEVEL 1 — RECALL

1. What is an airfoil by definition ●
   a. Any surface which generates a useful dynamic reaction of the air
   b. Only a wing surface which generates lift in forward flight
   c. Any surface which generates a useful static pressure of the air

2. Which two forces does an airfoil produce as the useful reaction for flying and manoeuvring ●
   a. Thrust and weight
   b. Lift and drag
   c. Lift and thrust

3. What is the leading edge of a rotor blade ●
   a. The edge that cuts into the wind when the rotor is turning
   b. The edge opposite the one that cuts into the wind
   c. The outer end of the blade opposite the root

4. What is the advancing blade ●
   a. The backward moving blade
   b. The forward moving blade
   c. The blade at the tail boom

5. What is the relative wind ●
   a. The direction of the airflow with respect to the ground
   b. The direction of the airflow with respect to the rotor blade
   c. The speed of the natural wind at the landing site

6. How is the relative wind directed in relation to the flight path of the blade ●
   a. Always along the same flight path
   b. Always opposite to the flight path
   c. Always perpendicular to the path

7. What is the axis of rotation ●
   a. The line from the leading edge to the trailing edge
   b. The plane formed by the average tip path
   c. The line about which the rotor blades turn

**8.** Which cockpit input changes the pitch angle of all main rotor blades ●

- **a.** The tail rotor pedals
- **b.** The collective pitch control
- **c.** The cyclic pitch stick

### LEVEL 2 — UNDERSTAND AND EXPLAIN

**9.** Why does a technician have to study the basic principles of helicopter flight ●

- **a.** To obtain the flight crew licence required for ground running of the rotor
- **b.** To understand the maintenance and troubleshooting practices on the systems
- **c.** To calculate the lift and thrust produced by the main rotor blades

**10.** Which surfaces act as the airfoils on a fixed-wing aircraft and on a helicopter ●

- **a.** Primarily the wings on the fixed-wing aircraft, the main rotor on the helicopter
- **b.** Primarily the fuselage on the fixed-wing aircraft, the tail rotor on the helicopter
- **c.** Primarily the propellers on the fixed-wing aircraft, the fuselage on the helicopter

**11.** Why is the helicopter often referred to as a rotary-wing aircraft ●

- **a.** Its fuselage acts as a rotating wing
- **b.** Its primary airfoil is the main rotor
- **c.** Its primary airfoil is the tail rotor

**12.** Which force produced by the main rotor blades holds the helicopter up against gravity? ●

- **a.** Drag
- **b.** Lift
- **c.** Thrust

**13.** A rotor blade is moving forward and downward through the air. In which direction does its relative wind act? ●

- **a.** Backward and downward
- **b.** Forward and upward
- **c.** Backward and upward

**14.** Why does the relative wind on a rotor blade change when a hovering helicopter begins to move horizontally ●

- **a.** Movement of the helicopter through the air is one of the factors that affect it
- **b.** The horizontal movement of the helicopter changes the span of the rotor blade
- **c.** The horizontal movement of the helicopter changes the chord line of the rotor blade

**15.** A helicopter stands with its rotor stopped while a strong natural wind passes across the blades. Which statement about the relative wind is correct? ●

- **a.** The relative wind is created only by blade flapping
- **b.** There is no relative wind because the rotor is stopped
- **c.** The wind blowing over the blade creates the relative wind

**16.** How can one cockpit control change both the pitch angle and the angle of attack of a rotor blade ●

- **a.** It tilts the plane of rotation, moving the relative wind against the chord
- **b.** It rotates the blade around the hub axis, moving the chord against each reference
- **c.** It lifts the whole blade, moving the axis of rotation against the chord

**17.** Which statement correctly distinguishes pitch angle from angle of attack? ●

- **a.** Pitch angle is referenced to the relative wind, angle of attack to the plane of rotation
- **b.** Pitch angle is referenced to the plane of rotation, angle of attack to the relative wind
- **c.** Pitch angle is referenced to the axis of rotation, angle of attack to the span line

**18.** Why do the same factors that affect the relative wind also affect the angle of attack ●

- **a.** Because the angle of attack depends upon the relative wind
- **b.** Because the angle of attack depends upon the blade span
- **c.** Because the relative wind depends upon the pitch angle

### LEVEL 3 — APPLY AND REASON

**19.** Two reference dimensions of a rotor blade are quoted, one measured along the blade centreline and one across the blade section. Which pair is correct? ●

- **a.** Span along the centreline, chord across the section
- **b.** Span along the centreline, root across the section
- **c.** Chord along the centreline, span across the section

**20.** A helicopter is in forward flight and one blade is on the advancing side. In which direction does the relative wind on that blade act? ●

- **a.** Backward and upward, across the blade movement
- **b.** Horizontally forward, with the blade movement
- **c.** Horizontally backward, opposite to the blade movement

**21.** What creates the relative wind on a helicopter hovering in completely still air ●

- **a.** The turning of the rotor blades
- **b.** The forward speed of the helicopter
- **c.** The wind speed and the wind direction

**22.** A blade is set so that its chord lies exactly along the average tip path of the blades. What is its pitch angle? ●

- **a.** Forty-five degrees, because the chord lies between the two references
- **b.** Ninety degrees, because the chord and the axis of rotation coincide
- **c.** Zero, because the chord and the plane of rotation coincide

**23.** The relative wind on a blade lies in the same direction as the plane of rotation and the pilot then increases the pitch angle. What happens to the angle of attack? ●

- **a.** It increases and becomes larger than the pitch angle
- **b.** It stays constant while the pitch angle increases
- **c.** It increases and stays equal to the pitch angle

**24.** Why can the angle of attack of a blade change while the collective pitch control is held still ●

- **a.** An outside influence has changed the plane of rotation
- **b.** An outside influence has changed the relative wind
- **c.** An outside influence has changed the chord of the blade

**25.** On a turning blade the relative wind does not lie in the direction of the plane of rotation. What follows for the pitch angle and the angle of attack? ●

- **a.** They cannot be equal to each other
- **b.** They must change at the same rate
- **c.** They must be equal to each other

## Answers — Terminology and blade geometry

1. **a** An airfoil is any surface that gives a useful dynamic reaction from the air it moves through. (L1, PDF p. 6)
2. **c** The text names lift and thrust as the two forces needed for flying and manoeuvring. (L1, PDF p. 6)
3. **a** The leading edge cuts into the wind. (L1, PDF p. 8)
4. **b** The blade moving forward is the advancing blade. (L1, PDF p. 10)
5. **b** Relative wind is airflow direction relative to the blade. (L1, PDF p. 12)
6. **b** The relative wind is always opposite to the blade's flight path. (L1, PDF p. 12)
7. **c** The rotor blades turn around the axis of rotation. (L1, PDF p. 16)
8. **b** The collective pitch control changes the pitch angle. (L1, PDF p. 18)
9. **b** Understanding the maintenance and troubleshooting practices carried out on the various systems is the reason a technician needs a thorough knowledge of the principles. (L2, PDF p. 4)
10. **a** The wings are the primary airfoil of a fixed-wing aircraft and the main rotor is the primary airfoil of the helicopter. (L2, PDF p. 4)
11. **b** Because its main lifting surface is a rotating rotor, the helicopter is called rotary-wing. (L2, PDF p. 4)
12. **b** Of the two forces the airfoil produces, lift is the one the manual gives as counteracting gravity. (L2, PDF p. 6)
13. **c** The relative wind is always exactly opposite to the flight path of the blade, so forward and downward movement gives a backward and upward relative wind. (L2, PDF p. 12)
14. **a** Horizontal movement of the helicopter is one of the factors that affect the relative wind. (L2, PDF p. 14)
15. **c** Even with the rotor stopped, wind over the blade is airflow relative to the blade. (L2, PDF p. 14)
16. **b** Both angles are measured from the chord, so rotating the blade about the hub axis changes both. (L2, PDF p. 18)
17. **b** Both use the chord, but pitch angle uses the plane of rotation and angle of attack uses the relative wind. (L2, PDF p. 18)
18. **a** Angle of attack is measured to the relative wind, so what changes one changes the other. (L2, PDF p. 20)
19. **a** Span runs root to tip along the centreline, and the chord runs leading edge to trailing edge across the section. (L3, PDF p. 6)
20. **c** The advancing blade moves forward, and the relative wind is always opposite to the blade's flight path. (L3, PDF p. 10)
21. **a** With no wind and no forward speed, only the turning of the blades is left to produce the relative wind. (L3, PDF p. 14)
22. **c** The pitch angle is measured between the chord and the plane of rotation, so a chord lying in that plane gives zero. (L3, PDF p. 16)
23. **c** With the relative wind in the plane of rotation the two angles are equal, and the angle of attack follows the pitch angle up. (L3, PDF p. 20)
24. **b** Angle of attack is measured from the relative wind, so it changes when outside influences move the relative wind. (L3, PDF p. 20)
25. **a** The two angles are equal only when the relative wind has the same direction as the plane of rotation, so with the relative wind elsewhere they cannot be equal. (L3, PDF p. 20)

# Lift and gyroscopic precession

PDF pages 22–24 · 9 questions

## LEVEL 1 — RECALL

**26.** What is lift ●

a. The force that is produced by the airfoil
b. The force that is produced by gravity
c. The force produced by torque reaction

**27.** How is gyroscopic precession defined ●

a. The turning of the fuselage opposite to the rotor
b. The change of the rotational axis as a result of an external force
c. The change of the rotor RPM as a result of an external force

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**28.** Why must gyroscopic precession be considered on the tail rotor as well as on the main rotor ●

a. Because the tail rotor blades have feathering hinges
b. Because the tail rotor turns opposite to the fuselage
c. Because the effect applies to every rotating body

**29.** An external force tries to turn a spinning wheel about its input axis. Around which axis does the movement actually take place? ●

a. Only around the spin axis
b. Around the output axis
c. Only around the input axis

**30.** Why must the steering input to the rotor be applied 90° before the point where the tilt is wanted ●

a. Because the swash plate is mounted 90° to the rotor
b. Because the output occurs 90° later in the direction of rotation
c. Because the output occurs 90° earlier in the direction of rotation

**31.** A downward force is applied on the right side of a counterclockwise turning rotor. Where does the output appear? ●

a. Right in front
b. Right at the rear
c. On the left side

## LEVEL 3 — APPLY AND REASON

**32.** What happens to the direction in which lift acts on a blade when the relative wind changes direction ●

a. It stays fixed, remaining perpendicular to the plane of rotation
b. It changes with it, staying parallel to the relative wind
c. It changes with it, staying perpendicular to the relative wind

**33.** What must happen to the plane of rotation when an external force tilts the axis of rotation ●

a. It tilts the opposite way, staying parallel to the axis
b. It stays level, while the axis tilts away from it
c. It tilts with the axis, staying perpendicular to it

**34.** A helicopter rotor turns counterclockwise. The tip-path plane must tilt backward for rearward flight. Following the precession rule, where must the downward force be put in? ●

- **a.** Right at the rear
- **b.** On the left side
- **c.** On the right side

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## Answers — Lift and gyroscopic precession

**26.** **a** Lift is the airfoil's force. (L1, PDF p. 22)

**27.** **b** Precession is a change of the rotational axis caused by an external force. (L1, PDF p. 24)

**28.** **c** Gyroscopic precession applies to all rotating bodies, so it applies to the tail rotor as well as the main rotor. (L2, PDF p. 24)

**29.** **b** Because of precession the movement occurs around the output axis. (L2, PDF p. 24)

**30.** **b** Precession shifts the result 90° in the direction of rotation, so the input must lead by 90°. (L2, PDF p. 24)

**31.** **a** 90° later in counterclockwise rotation from the right side is the front. (L2, PDF p. 24)

**32.** **c** Lift is defined as perpendicular to the relative wind, so its direction follows the relative wind. (L3, PDF p. 22)

**33.** **c** The plane of rotation is always perpendicular to the axis of rotation, so it tilts with the axis. (L3, PDF p. 24)

**34.** **b** The output is wanted at the rear; 90° earlier in counterclockwise rotation is the left side. (L3, PDF p. 24)

# Classification of rotary-wing aircraft

PDF pages 26–34 · 8 questions

## LEVEL 1 — RECALL

**35.** Which law explains torque reaction on a helicopter fuselage ●

- **a.** Newton's Second Law
- **b.** Newton's Third Law
- **c.** Newton's law of inertia

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**36.** Why must the torque of the main rotor be counteracted before flight is possible ●

- **a.** Because the fuselage would otherwise turn in the direction opposite to the rotor blades
- **b.** Because the main rotor would otherwise produce thrust instead of lift
- **c.** Because the rotor blades would otherwise turn in the direction opposite to the fuselage

**37.** A helicopter's main rotor turns counterclockwise seen from above and the tail rotor is not yet producing thrust. Which way does the fuselage tend to turn? ●

- **a.** Clockwise
- **b.** Not at all
- **c.** Counterclockwise

## LEVEL 3 — APPLY AND REASON

**38.** A pilot raises the collective lever and the turning force on the fuselage rises. Why must the tail rotor thrust then change too? ●

- **a.** Because it is what balances the lift of the main rotor
- **b.** Because it is what compensates the torque on the airframe
- **c.** Because it is what balances the drag of the main rotor

**39.** What do a gyrocopter in normal flight and a helicopter after an engine failure have in common

- **a.** The rotor is driven by the airflow in autorotation
- **b.** The rotor is driven by the engine through the drive shafts
- **c.** The rotor is held at a fixed pitch angle by the collective

**40.** The lift-producing rotor of a gyroplane stays connected to the engine in flight. What follows from that?

- **a.** The rotor has to be tilted slightly aft
- **b.** The rotor can no longer stall in flight
- **c.** The rotor torque has to be compensated

**41.** A convertiplane has accelerated to the airspeed at which its rotor turns freely in autorotation. What is then providing lift and what is providing thrust?

- **a.** The wings give the lift, the rotor gives the thrust
- **b.** The rotor gives the lift, the propellers or turbines give the thrust
- **c.** The wings give the lift, the propellers or turbines give the thrust

**42.** How does the rotor of a convertiplane in cruise differ from the rotor of a type 3 transformation helicopter in cruise

- **a.** Both rotors turn in autorotation and carry part of the aircraft weight
- **b.** The convertiplane rotor turns in autorotation, the type 3 rotor is shut down and stowed
- **c.** The type 3 rotor turns in autorotation, the convertiplane rotor is shut down and stowed

## Answers — Classification of rotary-wing aircraft

**35.** **b** Torque reaction is action and reaction, Newton's Third Law. (L1, PDF p. 26)

**36.** **a** Torque makes the fuselage rotate opposite to the blades, so it must be counteracted before flight. (L2, PDF p. 26)

**37.** **a** The fuselage reaction is opposite to a counterclockwise rotor, so clockwise. (L2, PDF p. 26)

**38.** **b** The tail rotor is what compensates the torque, so a change in torque has to be matched by a change in its thrust. (L3, PDF p. 26)

**39.** **a** Both have an unpowered rotor kept turning by the airflow, which is autorotation. (L3, PDF p. 28)

**40.** **c** An engine-driven rotor produces a torque reaction on the airframe, so it must be compensated. (L3, PDF p. 30)

**41.** **c** Once the rotor autorotates it makes no lift, so the wings carry the aircraft and the propellers or turbines push it. (L3, PDF p. 32)

**42.** **b** The convertiplane rotor keeps turning in autorotation while the type 3 rotor is deactivated and stowed. (L3, PDF p. 34)

# Cyclic, collective and anti-torque controls

PDF pages 36–44 · 20 questions

## LEVEL 1 — RECALL

**43.** What does lift counteract on a helicopter ●

a. Thrust
b. Torque
c. Weight

**44.** How do helicopters change direction, unlike other aircraft ●

a. By deflecting control surfaces on the fuselage
b. By changing the RPM of the main rotor
c. By changing the angle of the rotor blade plane

**45.** A pilot moves the cyclic stick in the direction of the nose. What does the rotor plane do ●

a. It tilts backward
b. It moves upward
c. It tilts forward

**46.** What does the helicopter do when the pilot moves the collective lever upwards ●

a. It turns left
b. It descends
c. It climbs

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**47.** What does a helicopter gain from being able to tilt its rotor plane slightly in any direction ●

a. Torque can act in all directions, giving heading control
b. Thrust can act in all directions, giving directional control
c. Lift can act in all directions, giving collective control

**48.** Why does a helicopter need no pull or push propellers

a. Because the rotors generate both lift and thrust
b. Because the rotors are driven by the airflow
c. Because the tail rotor generates the thrust

**49.** During an autorotation after an engine failure, which statement about control is correct? ●

a. Only the pedals remain effective and the touchdown point cannot be chosen
b. The helicopter is fully controllable and the pilot can choose the touchdown point
c. The helicopter is not controllable and lands wherever the descent takes it

**50.** Why does moving the collective lever also alter the torque effect on the fuselage ●

a. Because the movement of the lever also changes the engine power
b. Because the movement of the lever also tilts the rotor plane
c. Because the movement of the lever also changes the tail rotor pitch

**51.** A pilot wants to yaw the helicopter around the rotor mast axis. How do the pedals bring this about? ●

a. They change the pitch angle of the main rotor blades and so vary its thrust
b. They change the pitch angle of the tail rotor blades and so vary its thrust
c. They tilt the tail rotor plane and so vary the direction of its thrust

### LEVEL 3 — APPLY AND REASON

**52.** A helicopter's main rotor is driven by the engine. Which consequence follows for the fuselage ●

a. Its torque reaction on the fuselage has to be counteracted
b. Its rotor cannot be tilted out of the horizontal plane
c. Its downwash cannot escape from under the fuselage

**53.** How is a helicopter able to fly sideways ●

a. The blade pitch is increased on one side, so the lift acts sideways
b. The rotor plane is tilted sideways, so the thrust acts sideways
c. The tail rotor thrust is increased, so the fuselage is pushed sideways

**54.** How do the cyclic control and the collective control differ in the way they change the pitch angle of the main rotor blades ●

a. The cyclic changes it during each revolution, the collective changes all blades together
b. The collective changes it during each revolution, the cyclic changes all blades together
c. Both change it during each revolution, but only the cyclic changes all blades together

**55.** In terms of the law of acceleration, why must the rotor plane be tilted when changing from a hover into forward flight ●

a. A force is needed in the direction of flight, and thrust acts perpendicular to the rotor plane
b. A force is needed in the direction of flight, and thrust acts parallel to the rotor plane
c. A torque is needed in the direction of flight, and thrust acts perpendicular to the rotor mast

**56.** In straight-ahead flight the blade has minimum incidence in front and maximum incidence at the rear. What is the result for the rotor plane? ●

a. It tilts backward, giving lift and rearward thrust
b. It tilts forward, giving lift and forward thrust
c. It stays level, giving lift but no thrust

**57.** Why does raising the collective lever make the rotor produce more lift ●

a. The plane of rotation of every blade rises, and with it the relative wind
b. The pitch angle of every blade rises, and with it the angle of attack
c. The rotational speed of every blade rises, and with it the chord

**58.** Which hinge allows the pitch change commanded from the cockpit, and which allows the blade to move under aerodynamic forces ●

a. The flapping hinge for the pitch change, the feathering hinge for the movement
b. The lead-lag hinge for the pitch change, the flapping hinge for the movement
c. The feathering hinge for the pitch change, the flapping hinge for the movement

**59.** A pilot raises the collective lever to climb and the torque effect on the fuselage changes. Which control must the pilot then use to vary the thrust that compensates it? ●

a. The collective lever
b. The cyclic pitch stick
c. The tail rotor pedals

**60.** Why does varying the tail rotor thrust make the helicopter turn around the rotor mast axis ●

a. The thrust no longer exactly balances the torque reaction on the fuselage
b. The thrust tilts the main rotor disk away from the rotor mast axis
c. The thrust no longer exactly balances the lift produced by the main rotor

**61.** A pilot pushes the left pedal forward during a hover and holds the new heading. What else now needs attention? ●

- **a.** The lateral drift, because the tail rotor thrust has changed
- **b.** The pitch angle, because the collective lever has changed
- **c.** The rotor RPM, because the tail rotor pitch has changed

**62.** A pilot pushes the cyclic stick forward to fly forward. Which chain of events takes place at the swash plate and the blades? ●

- **a.** The swash plate turns, the bell cranks move and the blade pitch is held through the revolution
- **b.** The swash plate is lifted, the control rods move and the blade pitch rises on every blade at once
- **c.** The swash plate tilts, the control rods move and the blade pitch varies through the revolution

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## Answers — Cyclic, collective and anti-torque controls

**43.** **c** Lift counteracts weight. (L1, PDF p. 36)

**44.** **c** Helicopters tilt the rotor blade plane instead of deflecting surfaces. (L1, PDF p. 38)

**45.** **c** Forward cyclic tilts the rotor plane forward. (L1, PDF p. 38)

**46.** **c** More pitch gives more lift and the helicopter climbs. (L1, PDF p. 40)

**47.** **b** Tilting the rotor plane lets the thrust act in any direction, which is how directional control is obtained. (L2, PDF p. 36)

**48.** **a** The rotor provides both lift and thrust. (L2, PDF p. 36)

**49.** **b** During an autorotative landing the helicopter stays fully controllable and the pilot determines the location for touchdown. (L2, PDF p. 36)

**50.** **a** The collective lever changes engine power as well as blade pitch, and with it the torque. (L2, PDF p. 40)

**51.** **b** Pedal movement changes tail rotor blade pitch, which varies tail rotor thrust and yaws the aircraft. (L2, PDF p. 42)

**52.** **a** An engine driven rotor makes the fuselage tend to rotate the opposite way, and this torque has to be counteracted. (L3, PDF p. 36)

**53.** **b** Thrust acts perpendicular to the rotor plane, so tilting the plane sideways points the thrust sideways. (L3, PDF p. 36)

**54.** **a** Cyclic pitch varies around the disc each revolution while collective pitch moves all blades by the same amount. (L3, PDF p. 38)

**55.** **a** Accelerating forward needs a forward force, and the only way to point the thrust forward is to tilt the rotor plane. (L3, PDF p. 38)

**56.** **b** More lift at the rear and less at the front tilts the plane forward, producing lift and forward thrust. (L3, PDF p. 38)

**57.** **b** More pitch means more angle of attack on every blade, so the rotor makes more lift. (L3, PDF p. 40)

**58.** **c** Feathering hinges take the commanded pitch change, flapping hinges let the blade move under the air loads. (L3, PDF p. 40)

**59.** **c** Torque is compensated by the tail rotor, whose thrust the pilot varies with the pedals. (L3, PDF p. 40)

**60.** **a** The tail rotor thrust is what holds the torque reaction in balance, so changing it lets the fuselage turn. (L3, PDF p. 42)

**61.** **a** Tail rotor thrust is what produces the translating tendency, so changing it changes the drift the pilot must hold off. (L3, PDF p. 42)

**62.** **c** A cyclic input tilts the swash plate, which moves the control rods and varies blade pitch through each revolution. (L3, PDF p. 44)

# Swash plate, dissymmetry of lift and flapping

PDF pages 44–54 · 26 questions

## LEVEL 1 — RECALL

**63.** What is dissymmetry of lift ●

a. The difference in lift between advancing and retreating blade
b. The difference in lift between main and tail rotor
c. The difference in lift in and out of ground effect

**64.** Where does the blade's centre of gravity move when the blade flaps downwards ●

a. Closer to the blade tip
b. Closer to the rotational axis
c. Closer to the leading edge

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**65.** What acts on the stationary part of the swash plate, and what does the rotating part act on ●

a. Control rods and bell cranks from the cockpit act on it; it drives the blades through control rods
b. The rotor head acts on it through a bearing; it drives the bell cranks through control rods
c. Control rods from the blades act on it; it drives the cockpit controls through bell cranks

**66.** Why does the swash plate have to act as an interface ●

a. It transfers engine power between the main gearbox and the rotating rotor mast
b. It transfers control signals between the stationary input system and the rotating rotor
c. It transfers torque between the main rotor and the tail rotor drive system

**67.** A helicopter is hovering with no wind at all. Why is there no dissymmetry of lift in this condition? ●

a. There is no airstream from speed of flight, so both blades have the same parameters
b. There is no pitch angle on either blade, so both blades have the same parameters
c. There is no rotational speed at the blade tips, so both blades have the same parameters

**68.** Why does the advancing blade produce more lift in forward flight ●

a. The opposing airstream is subtracted from its rotational speed
b. Its pitch angle is increased by the collective
c. The opposing airstream is added to its rotational speed

**69.** Why does an upward flapping rotor blade accelerate while the rotor RPM stays constant ●

a. Its centre of gravity moves closer to the blade tip
b. Its centre of gravity moves closer to the rotational axis
c. Its centre of gravity moves closer to the trailing edge

**70.** A mass point on a rotating disk is moved towards the centre of rotation. What does it do to the rotational speed? ●

a. It has no effect on the rotational speed
b. It tries to accelerate the rotational speed
c. It tries to decelerate the rotational speed

**71.** Why does the advancing blade flap up ●

a. It experiences decreased relative wind and generates less lift
b. It experiences increased relative wind and generates more lift
c. Its centre of gravity moves closer to the blade tip

**72.** How far does blade flapping go towards restoring rotor balance ●

a. It helps to balance the rotor but is not enough alone
b. It increases the dissymmetry of lift instead
c. It restores the balance of the rotor completely

**73.** Why must a rotor blade be able to move and occupy a new position ●

a. Because the swash plate rotates with the rotor
b. Because aerodynamic forces act on the rotor blade
c. Because the torque reaction turns the fuselage

### LEVEL 3 — APPLY AND REASON

**74.** Why are the two parts of the swash plate held parallel but joined through a bearing ●

a. So the rotor plane can be tilted without moving the control rods above it
b. So the cockpit input can be passed to blades that are turning with the rotor
c. So the cockpit input can be passed to blades that are fixed to the rotor mast

**75.** During a ground check the whole swash plate is seen to rise without tilting, and every blade takes a larger pitch angle. What will the helicopter do if this is done in flight? ●

a. It flies forward, because the rotor plane is tilted forward
b. It climbs, because all blades gain pitch together
c. It yaws left, because the tail rotor pitch is increased

**76.** Which cockpit control changes a blade pitch angle without acting through the main rotor swash plate ●

a. The collective lever, which changes the main rotor blade pitch
b. The cyclic stick, which changes the main rotor blade pitch
c. The pedals, which change the tail rotor blade pitch

**77.** A technician watches the swash plate while a helper moves one cockpit control, and the whole swash plate moves up without tilting. Which control is being moved? ●

a. The cyclic stick
b. The tail pedals
c. The collective lever

**78.** During a control check the swash plate tilts but its mean height does not change. Which pitch change is being made? ●

a. Collective pitch change
b. Cyclic pitch change
c. Tail rotor pitch change

**79.** Where on the rotor disc does the highest airspeed occur when a helicopter flies forward ●

a. At the root of the advancing blade
b. At the tip of the retreating blade
c. At the tip of the advancing blade

**80.** A helicopter hovers on the spot in a steady strong wind. Why can its blades not all produce the same lift? ●

a. The wind adds to the speed of one half of the disc and subtracts from the other
b. The wind raises the pitch angle on one half of the disc and lowers it on the other
c. The wind increases the rotational speed of the whole rotor disc

**81.** A blade flaps up and therefore accelerates and moves forward in the plane of rotation. Which hinge has to allow this? ●

- **a.** The lead-lag hinge
- **b.** The feathering hinge
- **c.** The flapping hinge

**82.** In forward flight the retreating blade flaps down. Combining the flapping and centre-of-mass facts, how does this blade move in the plane of rotation? ●

- **a.** It accelerates and moves forward (leads)
- **b.** It keeps its position in the plane
- **c.** It decelerates and moves backward (lags)

**83.** Why is blade flapping a self-correcting effect ●

- **a.** The blade with more lift flaps down, which reduces its angle of attack and its lift
- **b.** The blade with more lift flaps up, which reduces its pitch angle and its lift
- **c.** The blade with more lift flaps up, which reduces its angle of attack and its lift

**84.** On which blade does flapping increase the angle of attack, and why must it do so ●

- **a.** The retreating blade, because it is the one producing too much lift
- **b.** The retreating blade, because it is the one producing too little lift
- **c.** The advancing blade, because it is the one producing too little lift

**85.** Why does upward flapping reduce the angle of attack of the advancing blade? ●

- **a.** Moving up reduces the blade pitch angle
- **b.** Moving up adds an upward relative wind component
- **c.** Moving up adds a downward relative wind component

**86.** A rotor head is built with no flapping hinges at all. How can dissymmetry of lift still be balanced on this rotor? ●

- **a.** The blades themselves flex and absorb the forces acting on them
- **b.** The feathering hinges let the blades change pitch instead
- **c.** The lead-lag hinges let the blades move up and down instead

**87.** A helicopter is in steady forward flight and the pilot makes no control input at all. Is there still lead-lag movement, and why? ●

- **a.** Yes, because dissymmetry of lift keeps the blades flapping
- **b.** Yes, because the collective lever holds the blades at a fixed pitch
- **c.** No, because lead-lag only follows a cyclic control input

**88.** A pilot in forward flight moves the cyclic stick to the right. Which blade movement is produced in addition to the pitch change? ●

- **a.** A lead-lag movement, because the rotor RPM changes
- **b.** A lead-lag movement, because the rotor plane tilts
- **c.** A feathering movement, because the rotor plane tilts

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### Answers — Swash plate, dissymmetry of lift and flapping

**63.** **a** The lift difference between advancing and retreating blade is dissymmetry of lift. (L1, PDF p. 46)

**64.** **a** Downward flapping moves the CG outward toward the tip. (L1, PDF p. 48)

**65.** **a** The cockpit acts on the stationary part and the rotating part is connected onward to the blades. (L2, PDF p. 44)

**66.** **b** Cockpit inputs are stationary but the blades rotate, so the swash plate links the two. (L2, PDF p. 44)

67. **a** Without the airstream caused by the speed of flight there is nothing to add to one blade and subtract from the other. (L2, PDF p. 46)
68. **c** The airstream adds to its speed, giving more lift. (L2, PDF p. 46)
69. **b** A mass moved towards the centre of rotation tries to speed the rotation up, so the up-flapping blade accelerates. (L2, PDF p. 48)
70. **b** Moving inward, it keeps its higher speed and so accelerates the rotation. (L2, PDF p. 48)
71. **b** More relative wind means more lift, so it flaps up. (L2, PDF p. 50)
72. **a** Flapping helps but does not fully restore balance. (L2, PDF p. 50)
73. **b** Aerodynamic forces require the blade to move. (L2, PDF p. 52)
74. **b** The bearing lets the upper part turn with the rotor while the parallel lower part stays still, passing the input across. (L3, PDF p. 44)
75. **b** Lifting the swash plate is a collective pitch increase, which makes more lift and a climb. (L3, PDF p. 44)
76. **c** The swash plate serves the main rotor blades, while the pedals act on the tail rotor blades. (L3, PDF p. 44)
77. **c** Lifting the swash plate is collective; tilting would be cyclic. (L3, PDF p. 44)
78. **b** Tilting the swash plate produces cyclic pitch change. (L3, PDF p. 44)
79. **c** Speed rises towards the tip and the opposing airstream adds on the advancing side, so the advancing tip is fastest. (L3, PDF p. 46)
80. **a** Wind is a relative wind just like forward speed, so it adds on the advancing side and subtracts on the retreating side. (L3, PDF p. 46)
81. **a** Movement forward and backward in the plane of rotation is taken by the lead-lag hinge. (L3, PDF p. 48)
82. **c** The retreating blade flaps down, its CG moves outward, so it decelerates and lags. (L3, PDF p. 48)
83. **c** The extra lift causes the up-flap, and the up-flap takes the angle of attack and the lift back down again. (L3, PDF p. 50)
84. **b** The retreating blade is short of lift, and flapping down raises its angle of attack and so its lift. (L3, PDF p. 50)
85. **c** Relative wind is opposite to blade motion, so an upward-moving blade meets air from above, reducing the angle of attack. (L3, PDF p. 50)
86. **a** Where there is no flapping hinge, the blades must instead have properties that allow them to absorb the forces. (L3, PDF p. 52)
87. **a** Dissymmetry of lift makes the blades flap in forward flight even with no input, and flapping always makes lead-lag. (L3, PDF p. 54)
88. **b** A cyclic input tilts the rotor plane and the tilt generates a lead-lag movement. (L3, PDF p. 54)

# Translating tendency, ground effect and autorotation

PDF pages 56–60 · 17 questions

## LEVEL 1 — RECALL

**89.** Which turning force does a counterclockwise turning main rotor generate on the fuselage ●

a. A clockwise turning force
b. A nose-down pitching force
c. A counterclockwise turning force

**90.** During powered flight, how is the rotor drag overcome ●

a. With rotor inertia
b. With upward airflow
c. With engine power

**91.** In which two failure cases must some other force sustain the rotor RPM ●

a. Engine failure or broken drive shafts
b. Tail rotor failure or broken pedals
c. Swash plate failure or broken control rods

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**92.** A helicopter is in a steady hover. Which two jobs is the thrust of the tail rotor doing at the same time? ●

a. Compensating the main rotor torque and preventing the lateral drift
b. Maintaining heading control and tilting the main rotor disk slightly
c. Compensating the main rotor torque and maintaining heading control

**93.** A helicopter in hover drifts sideways although torque is compensated. What causes this drift? ●

a. The thrust of the tail rotor
b. Dissymmetry of lift
c. Gyroscopic precession

**94.** Why does ground effect become more pronounced the nearer the ground is approached ●

a. Because it is caused by the surface reducing the rotational speed of the blades
b. Because it is caused by the surface increasing the pitch angle of the blades
c. Because it is caused by the surface interfering with the rotor airflow pattern

**95.** Why must a helicopter in autorotation descend ●

a. Because the collective pitch is lowered to its minimum stop during autorotation
b. Because the airflow up through the rotor in the descent supplies the energy
c. Because the engine power is needed to overcome the drag of the rotor blades

## LEVEL 3 — APPLY AND REASON

**96.** Which cockpit control does the pilot use to hold off the drift caused by translating tendency ●

a. The collective lever, which lifts the main rotor disk
b. The pedals, which tilt the tail rotor thrust line
c. The cyclic stick, which tilts the main rotor disk

**97.** How does tilting the main rotor disk stop the lateral drift of a helicopter in the hover ●

a. The thrust then acts partly against the rotor mast axis
b. The lift then acts partly against the torque direction
c. The thrust then acts partly in the opposite sideways direction

**98.** During hover, tail rotor thrust makes the helicopter drift to the right. How must the main rotor disk be tilted to prevent this? ●

- **a.** Slightly to the right
- **b.** Slightly forward
- **c.** Slightly to the left

**99.** Why does the reduced downward airflow in ground effect increase the angle of attack of the blades ●

- **a.** The pitch angle changes, and the angle of attack is measured from it
- **b.** The relative wind direction changes, and the angle of attack is measured from it
- **c.** The plane of rotation changes, and the angle of attack is measured from it

**100.** A helicopter settles into a hover close to the ground and the pilot wants to stay there. Which lever does he move and what happens to the power needed? ●

- **a.** He lowers the collective, and more power is needed
- **b.** He raises the collective, and more power is needed
- **c.** He lowers the collective, and less power is needed

**101.** A helicopter hovering just above the ground climbs slowly until it is just under one rotor diameter up. What has happened to ground effect? ●

- **a.** It is stronger and still present
- **b.** It is lost once the skids leave the ground
- **c.** It is weaker but still present

**102.** In ground effect the lift vector of the blades becomes more vertical. Why does that reduce the power the pilot needs? ●

- **a.** More of the rotor force acts straight against the weight
- **b.** More of the rotor force acts straight along the blade span
- **c.** More of the rotor force acts straight against the drag

**103.** Once the engine has failed, what does the collective pitch lever control ●

- **a.** The engine power that keeps the rotor turning in the descent
- **b.** The tail rotor thrust that keeps the heading in the descent
- **c.** The force that keeps the rotor turning in the descent

**104.** An engine fails in cruise and the pilot chooses a field to land in. What makes that choice possible? ●

- **a.** The rotor keeps turning at an RPM that provides aircraft control
- **b.** The rotor keeps turning at an RPM that provides enough lift to hover
- **c.** The engine keeps turning at an RPM that provides hydraulic power

**105.** An engine fails in cruise. In which order does the pilot use energy during the autorotation? ●

- **a.** Altitude drives the rotor in the descent, then rotor kinetic energy cushions touchdown
- **b.** Rotor kinetic energy cushions the descent, then altitude drives the rotor at touchdown
- **c.** Engine power drives the rotor in the descent, then altitude cushions touchdown

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## Answers — Translating tendency, ground effect and autorotation

**89.** **a** The fuselage force is opposite, so clockwise. (L1, PDF p. 56)

**90.** **c** In powered flight, engine power overcomes rotor drag. (L1, PDF p. 60)

**91.** **a** Engine failure or broken drive shafts require another force. (L1, PDF p. 60)

**92.** **c** In the hover the tail rotor thrust both compensates the main rotor torque and maintains heading control; the drift it causes is held off with the main rotor disk instead. (L2, PDF p. 56)

**93.** **a** The tail rotor's sideways thrust causes translating tendency. (L2, PDF p. 56)

**94.** **c** The effect comes from the surface interfering with the rotor airflow, so it grows as the surface gets closer. (L2, PDF p. 58)

**95.** **b** The upward airflow that keeps the rotor turning only exists while the helicopter is descending. (L2, PDF p. 60)

**96.** **c** Tilting the rotor disk is a cyclic input, so the cyclic stick is the control that holds off the drift. (L3, PDF p. 56)

**97.** **c** Thrust acts perpendicular to the rotor plane, so tilting the disk the other way points some thrust against the drift. (L3, PDF p. 56)

**98.** **c** The disk is tilted opposite to the drift, so to the left for a right drift. (L3, PDF p. 56)

**99.** **b** A slower downward airflow changes the relative wind, and the angle of attack is taken from the relative wind. (L3, PDF p. 58)

**100.** **c** Reducing blade pitch is done with the collective, and hovering in ground effect needs less power. (L3, PDF p. 58)

**101.** **c** Ground effect reaches to one rotor diameter and weakens with height, so it is still there but reduced. (L3, PDF p. 58)

**102.** **a** Lift is what overcomes weight, so a more vertical lift vector puts more of the rotor force where it is needed. (L3, PDF p. 58)

**103.** **c** With no engine the collective no longer sets engine power; it controls the force that sustains rotor RPM. (L3, PDF p. 60)

**104.** **a** Autorotation keeps the rotor at a controlling RPM, so the helicopter stays fully controllable and the pilot picks the spot. (L3, PDF p. 60)

**105.** **a** Phase 1 converts altitude into rotor energy; phase 2 uses the rotor energy for touchdown. (L3, PDF p. 60)